Kernel Affine Hull Features for Time-Series Forecasting: Distinctness, Bounded Corrections, and Predictive Limits*

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Abstract

Kernel Affine Hull Machines (KAHMs) reconstruct observations from local reference sets. We study whether these reconstructions provide useful features for a finite-dimensional forecasting decoder. Direct transfer of normalized KAHM weights to reference targets is a ratio of kernel-ridge predictions. We give sufficient conditions under which KAHM and matched radial-basis-function (RBF) reconstructions generate the same augmented feature span, and construct an irregular reference set for which the spans differ. Separately, bounded features and a ridge residual decoder yield a uniform bound on the forecast correction, including its free intercept. A prospectively specified experiment compares ten procedures on 192 task suites: three nonlinear simulation laws and a linear control, two observation-noise levels, and 24 independent repetitions. The bounded KAHM departure improves aggregate nonlinear normalized mean squared error by 12.58% over common local-coordinate features. Its gains over matched RBF reconstruction and raw-input RBF forecasting are 1.17% and 3.92%; neither establishes the prespecified 3% margin under simultaneous inference. Boosted trees and a validation-selected non-KAHM procedure attain lower aggregate error. The correction bounds and linear-control noninferiority criterion are satisfied. Supporting studies identify conditional gains, unfavorable natural-data comparisons, and limitations of adaptation. Reconstruction can therefore supply useful forecasting features, but algebraic distinctness and bounded corrections do not establish a practical advantage specific to KAHM.

1 Introduction

A reconstruction of an observed history and a prediction of its future solve different problems. Kernel Affine Hull Machines (KAHMs) construct nonlinear reconstruction maps from local reference observations (Kumar et al., 2024). A small reconstruction error does not establish that the resulting representation is useful for forecasting. For example, a reconstruction residual that vanishes along an affine line loses position along that line, although the future target may vary with position. Conversely, a deterministic feature cannot add information to the history from which it is computed, but it can make useful functions available to a restricted decoder.

This distinction raises three questions. Does the reconstruction supply functions that a simpler matched representation cannot express? Can the magnitude of a fitted forecast correction be controlled? Does the resulting procedure improve prediction at the available sample size and computational budget? Each question requires its own evidence. A comparison of function spans concerns algebraic expressiveness; a correction bound concerns the size of one prediction component; a forecasting comparison concerns the complete fitted and selected procedure.

We investigate these questions through *reconstruction departures*. For each local model, a departure is the difference between its nonlinear reconstruction and its affine projection, multiplied by a locality weight.

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The decoder also receives the locality weights and weighted principal coordinates. These common features retain principal-coordinate position within each local model and are shared by all reconstruction comparisons. Replacing the nonlinear map with its affine projection makes the raw departure zero. A positive-kernel RBF reconstruction, using the same references, kernel, and locality weights, provides a closely matched nonlinear alternative. A residual decoder then adds a calibrated correction to a fixed raw-history ridge forecast.

The algebraic analysis identifies both redundancy and distinctness. Normalized transfer of KAHM reconstruction weights to reference targets is exactly a ratio of two kernel-ridge predictions. Under a constant-row-sum and invariant-subspace condition, KAHM and RBF departures have the same augmented raw feature span. An irregular three-reference construction shows that the spans can also differ. These results concern untransformed functions. They do not equate fitted ridge predictions under different parameterizations, nor do they establish an advantage for every reference bank or target.

The second contribution is a uniform magnitude bound for the residual correction. Each feature block is standardized on the fitting set and mapped into a bounded ball before ridge regression is fitted with an unpenalized intercept. The coefficient and correction bounds account for the empirical residual mean and the scaling of the penalty. They apply to bounded features generally and therefore support the specified KAHM construction without implying a property unique to KAHM. In particular, they do not bound the raw baseline or the total forecast error.

The principal experiment compares ten procedures over four simulation laws, two added-noise levels, and 24 independent repetitions. Fitting, hyperparameter selection, correction calibration, and final evaluation use separate episode sets. The protocol and prepared inputs were recorded locally before the final forecasts were computed; this was not a public preregistration. Five simultaneous comparisons test a 3% nonlinear-loss improvement margin, and a sixth tests noninferiority to ridge on a linear control with a 5% tolerance.

The bounded KAHM departure improves on common local-coordinate features, but the full prespecified improvement criterion is not satisfied. The gain over matched RBF departures is below the required margin, and the margin over raw RBF forecasting remains unresolved. Boosted trees and the validation-selected non-KAHM procedure have lower aggregate loss. Supporting controlled, natural-data, and adaptive studies show why this distinction matters beyond the principal experiment. Their findings are summarized in Section 7; their complete methods and results appear in the companion *Supplementary Studies*.

The paper thus contributes a characterization of the reconstruction feature space, a correction bound for a specified residual decoder, and an empirical evaluation that separates a useful reconstruction feature from an established advantage specific to KAHM. Full proofs of the central results and the complete principal experimental protocol are included in Appendices A and B.

2 Related work

Local geometry and reconstruction. Prediction from delay coordinates using local approximation predates the present representations (Farmer & Sidorowich, 1987). KAHMs were introduced for geometrically informed classification, including a differential-privacy application (Kumar et al., 2024), and developed for collaborative learning (Kumar et al., 2025). Those task results do not establish forecasting utility. Local coordinate coding connects localized representations with approximation of smooth functions (Yu et al., 2009). Local tangent coding adds weighted principal-direction displacements (Yu & Zhang, 2010); these are the common features retained in our matched comparisons. With linear readouts they are also related to local affine experts (Jacobs et al., 1991). Our question is whether the additional KAHM reconstruction departures change the available functions and improve a specified forecasting procedure beyond these common features.

Kernel regression and predictive representations. The matched positive-kernel reconstruction is a vector-valued kernel smoother that averages reference coordinates with normalized Gaussian weights (Nadaraya, 1964). The target-transfer identity in Proposition 1 relates KAHM to kernel ridge regression, while Propositions 2 and 3 characterize equality and inequality of the corresponding augmented raw feature spans. Reconstruction and forecasting objectives also occur in learned dynamical representations: Lusch et al. (2018) combine reconstruction, latent linearity, and multistep prediction, whereas Liu et al. (2023) optimize a

residual Koopman forecaster without the canonical reconstruction loss. Here the principal experiment fixes the geometric bank and compares its departures with a matched RBF reconstruction under identical feature transformations and a restricted residual decoder.

Information, regularization, and residual fitting. Supervised dimension reduction studies preservation of the input–target relationship (Fukumizu et al., 2004), and predictive $\mathcal{V}$-information distinguishes information accessible to a restricted observer (Xu et al., 2020). Our feature-space analysis addresses a finite decoder and does not estimate unrestricted Bayes risk on natural data. Residual fitting under squared loss is an established construction (Bühlmann & Yu, 2003). The correction bound in Proposition 4 follows from the singular values of centered ridge regression and is not specific to KAHM. Nyström approximation changes the available coordinates and computational cost (Williams & Seeger, 2001; Rudi et al., 2015); the raw RBF comparator is therefore evaluated as a complete fitted procedure, not as a presumed equivalent function class.

Evaluation and relation to prediction-limit studies. Model-selection bias (Cawley & Talbot, 2010) and forecast-evaluation pitfalls (Hewamalage et al., 2023) motivate distinct fitting, selection, calibration, and evaluation roles. The supporting natural-data studies use the Monash archive (Godahewa et al., 2021) and are reported separately from the principal simulated experiment. Related work by Kumar & Kargaran (2026) studies successor soft coordinates, norm-bounded prediction certificates, and Koopman closure. The present study shares KAHM source components and some generator definitions with that work, but evaluates particular finite forecasting procedures against matched alternatives. Its reconstruction-span results and correction bound are established under their own assumptions. The paired-error intervals and measured costs do not provide class-wide impossibility results or minimum-cost certificates.

3 Forecasting features from local reconstruction

Let (y_t) be a scalar observed series. At forecast origin t, the available history is $x_t = (y_{t-d+1}, \dots, y_t)^\top \in \mathbb{R}^d$, and the target at horizon $h \geq 1$ is y_{t+h}. A forecasting procedure maps the history to a separate prediction for each h in a specified finite horizon set $\mathcal{H}$. The representation and decoder are fitted using a set A of history–target pairs. Distinct episode sets B, C, and D are used for model selection, correction calibration, and final evaluation, respectively; Section 6 specifies their construction.

Vectors are columns. The notation $v^\top$ denotes transpose, $\|v\|_2$ the Euclidean norm, and $\|v\|_1$ the sum of absolute coordinates. The matrix I_p is the $p \times p$ identity, and $\mathbf{1}_n$ is the length-n vector of ones; omitted subscripts are determined by the displayed products. Expectations and variances refer to the probability law specified for each statement or experiment. All feature-space statements condition on a fixed fitted reference bank.

3.1 Reference banks and reconstruction maps

Let $z \in \mathbb{R}^d$ be the coordinatewise standardized version of a history x, using means and scales estimated on A. A local reference bank is a collection of fitted models indexed by j. Model j has n_j reference histories stored as the rows of $Q_j \in \mathbb{R}^{n_j \times d}$, center $\mu_j \in \mathbb{R}^d$, and orthonormal principal directions $P_j \in \mathbb{R}^{d \times r_j}$. Here r_j is its local rank and $P_j^\top P_j = I_{r_j}$. Write $q_{ji} \in \mathbb{R}^d$ for reference i, so row i of Q_j is $q_{ji}^\top$. Let S_j be the sample covariance matrix of the projected references $P_j^\top q_{ji}$. For positive rank and positive-definite S_j, the shared Gaussian kernel is

$$\kappa_j(z, z') = \exp\left\{ -\frac{[P_j^\top (z - z')]^\top S_j^{-1} [P_j^\top (z - z')]}{2r_j} \right\}.$$

Define the kernel vector $k_j(z) \in \mathbb{R}^{n_j}$ by $[k_j(z)]_i = \kappa_j(z, q_{ji})$, and the Gram matrix $K_j \in \mathbb{R}^{n_j \times n_j}$ by $[K_j]_{i\ell} = \kappa_j(q_{ji}, q_{j\ell})$. With a positive reconstruction regularizer λ_j, set $C_j = (K_j + \lambda_j I_{n_j})^{-1}$. The Gaussian Gram matrix and its regularized inverse C_j are symmetric. The regularizer is fitted from A by the reference-reconstruction rule specified in Appendix B.

The KAHM and matched positive-kernel reconstructions are

$$A_{K,j}(z) = \frac{Q_j^\top C_j k_j(z)}{\mathbf{1}^\top C_j k_j(z)}, \qquad A_{R,j}(z) = \frac{Q_j^\top k_j(z)}{\mathbf{1}^\top k_j(z)}. \tag{1}$$

Both take values in $\mathbb{R}^d$. The first is defined when its signed denominator is nonzero; the second has a positive denominator for finite inputs. The matched affine reconstruction is $A_{A,j}(z) = \mu_j + P_j P_j^\top (z - \mu_j)$.

3.2 Common features and reconstruction departures

Models are grouped into partitions. If $\mathcal{J}_b$ indexes the models in partition b, their shared locality weights are

$$g_j(z) = \frac{\exp\{-\|z - \mu_j\|_2^2/(2s_j)\}}{\sum_{\ell \in \mathcal{J}_b} \exp\{-\|z - \mu_\ell\|_2^2/(2s_\ell)\}}, \qquad j \in \mathcal{J}_b,$$

where $s_j > 0$ is the fitted local squared-radius scale. These weights, called gates, sum to one within each partition and do not depend on the reconstruction being compared.

The common feature vector $\psi_0(z)$ contains all gates $g_j(z)$ and all weighted principal-coordinate vectors $g_j(z)P_j^\top(z - \mu_j)$. Define the reconstruction departures by

$$\begin{aligned} d_K(z) &= \mathrm{concat}_j\big[g_j(z)\{A_{K,j}(z) - A_{A,j}(z)\}\big], \\ d_R(z) &= \mathrm{concat}_j\big[g_j(z)\{A_{R,j}(z) - A_{A,j}(z)\}\big]. \end{aligned} \tag{2}$$

The operator concat stacks its column-vector arguments in a fixed order. Each departure retains all history coordinates, including components outside the local principal subspace. Substituting the affine map for the nonlinear reconstruction makes the departure identically zero. The common-feature procedure is therefore the nested control without reconstruction departures. The RBF departure is matched to the KAHM departure in references, kernels, gates, and feature dimension.

The affine control is nested at the level of the raw feature construction. The comparison with positive-kernel RBF reconstruction additionally holds fixed the number and locations of references, the projected kernel, the locality weights, and the decoder architecture. The difference between the two raw departure blocks is therefore the use of C_j in the normalized reconstruction. This matching does not assert equality of fitted predictions, and it does not compare KAHM with every possible kernel or nonlinear decoder.

3.3 Defined evaluation of signed normalizations

To guard against cancellation in the signed KAHM normalization, define

$$a_j(z) = C_j k_j(z), \qquad \rho_j(z) = \frac{|\mathbf{1}^\top a_j(z)|}{\|a_j(z)\|_1} \quad \text{when } \|a_j(z)\|_1 > 0. \tag{3}$$

The ratio is invariant to a common positive rescaling of the query kernel. When $\rho_j(z) \le 10^{-10}$, or a required quantity is undefined or nonfinite, the local reconstruction is replaced by its affine map, making that raw departure zero. The same finite-value checks apply to the positive RBF map. All affected rows are recorded. This fallback gives a defined feature vector where the original ratio cannot be evaluated reliably; it does not establish continuity across the fallback boundary.

A zero raw departure under fallback need not remain zero after A-fitted feature centering. The fallback defines the input to the subsequent transformation; the norm bound in Section 5 applies to that transformed input. The algebraic propositions in Section 4 concern the exact ratios on their stated domains, rather than the piecewise numerical fallback procedure.

4 Normalized reconstruction and feature spans

The reconstruction formulas permit two different questions: whether a particular map is an established kernel predictor, and whether its coordinates give a linear decoder functions unavailable from the matched

reconstruction. We answer these questions for fixed reference geometry. The results concern exact functions before feature standardization, radial transformation, or penalized fitting.

Consider one local model from Section 3 and suppress its index. Write M for its number of references, $Q \in \mathbb{R}^{M\times d}$ for their row-stored coordinates, and $C = (K + \lambda I_M)^{-1}$ for its regularized inverse, with $\lambda > 0$. The matrix K is the symmetric positive-semidefinite Gaussian Gram matrix. The vector $\mathbf{1}_M$ contains M ones, and $k(z) \in \mathbb{R}^M$ contains the kernel similarities of a query z to the references. Let A_K, A_R, and A_A denote the KAHM, positive-kernel RBF, and affine reconstructions in Equation (1).

4.1 The meaning of normalized target transfer

Proposition 1 (Normalization and target transfer)**.** *For a query with $s(z) = \mathbf{1}_M^\top Ck(z) \neq 0$, define $\omega(z) = Ck(z)/s(z)$. Then $\mathbf{1}_M^\top \omega(z) = 1$, and for every vector of reference targets $y \in \mathbb{R}^M$,*

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I_M)^{-1} y}{k(z)^\top (K + \lambda I_M)^{-1} \mathbf{1}_M}. \tag{4}$$

This is the ratio of two kernel-ridge predictions with regularization matrix $K + \lambda I_M$ and no additional intercept. For an affine scalar function $f(q) = q^\top b + c$, where $b \in \mathbb{R}^d$ and $c \in \mathbb{R}$, its reference values $f(Q) = Qb + c\mathbf{1}_M$ satisfy

$$\omega(z)^\top f(Q) - f(z) = \{A_K(z) - z\}^\top b.$$

The proof uses symmetry of C and the fact that the normalized weights sum to one. The numerator in Equation (4) fits the supplied targets, while the denominator fits the constant target. Because the denominator varies with the query, the ratio generally differs from the ordinary unnormalized kernel-ridge predictor. The affine identity also specifies the effect of reconstruction error on an affine target: only its component in the target direction b enters. The complete proof is in Appendix A.

4.2 Equal spans under reference symmetry

For a fixed common gate $g(z)$, define $d_K(z) = g(z)\{A_K(z) - A_A(z)\}$ and $d_R(z) = g(z)\{A_R(z) - A_A(z)\}$. Let $\mathcal{S}_0$ be the linear span of the scalar gate and the coordinates of $g(z)P^\top(z - \mu)$, where μ and P are the reference center and principal directions. The coordinates of $g(z)A_A(z) = g(z)\mu + P\{g(z)P^\top(z - \mu)\}$ belong to $\mathcal{S}_0$. Adjoining a vector-valued feature to a span means adjoining each of its scalar coordinate functions. A common intercept may be adjoined to both spans without changing the equality below.

Proposition 2 (A sufficient condition for equal feature spans)**.** *Suppose $K\mathbf{1}_M = \kappa\mathbf{1}_M$ for a scalar κ, and C leaves $\mathcal{V} = \operatorname{span}\{\mathbf{1}_M, Q_{:1}, \ldots, Q_{:d}\}$ invariant. Here $Q_{:i}$ is column i of Q, and invariance means $C\mathcal{V} \subseteq \mathcal{V}$. Then*

$$\operatorname{span}(\mathcal{S}_0, d_K) = \operatorname{span}(\mathcal{S}_0, d_R).$$

More explicitly, there are $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$ such that, with $a = (\kappa + \lambda)^{-1} > 0$,

$$C\mathbf{1}_M = a\mathbf{1}_M, \qquad CQ = QB + \mathbf{1}_M c^\top, \qquad A_K(z) = \{B^\top A_R(z) + c\}/a.$$

Invariance supplies the displayed representation of CQ. Substitution in the normalized reconstruction expresses each KAHM departure using RBF departures and the common affine features. The restriction of C to the finite-dimensional space $\mathcal{V}$ is invertible onto that space, which gives the reverse inclusion. This argument also covers redundant reference-coordinate columns; it does not require B to be invertible. Appendix A gives the full proof.

A two-reference example makes the role of the common coordinates explicit. Take scalar references $(-1, 1)$, $g = 1$, $P = 1$, and $A_A(z) = z$. If ρ is the off-diagonal Gram entry, then

$$A_K = c_\rho A_R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1 + \rho + \lambda}{1 - \rho + \lambda} > 0.$$

The departures differ, but adjoining the coordinate z makes their augmented spans equal. Thus a nonlinear reconstruction formula alone does not establish an additional raw basis.

4.3 An irregular reference bank gives a distinct basis

Proposition 3 (Irregular scalar references)**.** *Let $q_1 < q_2 < q_3$ satisfy $q_1 + q_3 \neq 2q_2$, and let $k_i(z) = \exp\{-t(z-q_i)^2/2\}$ for a precision $t > 0$. Set $K_{ij} = \exp\{-t(q_i - q_j)^2/2\}$ and $C = (K + \lambda I_3)^{-1}$ for $\lambda > 0$. Suppose every entry of $C\mathbf{1}_3$ is positive. With $Q = (q_1, q_2, q_3)^\top$, $g = 1$, and $A_A(z) = z$,*

$$A_K \notin \operatorname{span}(1, z, A_R), \qquad d_K \notin \operatorname{span}(1, z, d_R).$$

These spans consist of scalar functions on the whole real line.

The six products of kernel functions have distinct exponential factors because their exponents contain the six distinct sums $q_i + q_j$ for $i \leq j$. Their linear independence rules out an affine relation between A_K and A_R: such a relation would force K to have constant row sums, whereas its middle row has a strictly larger sum than either outer row. Positivity of $C\mathbf{1}_3$ makes both reconstructions bounded, excluding an additional nonzero multiple of z. Appendix A provides every step of this argument.

For the explicit references $(-1, 0, 2)$, the sample-variance precision is $t = 3/7$. The reference-based regularization rule gives $\lambda = 4.498274743339644$, for which positivity of $C\mathbf{1}_3$ follows from the matrix bound in the appendix. This supplies a concrete example satisfying the proposition. It does not assert distinctness for every higher-dimensional bank fitted in the forecasting experiment.

Together, the propositions delimit the algebraic contribution. Some reference geometries yield equal augmented spans, while an irregular geometry can supply a distinct function. Neither conclusion determines forecasting accuracy. A general change of feature coordinates changes an isotropic ridge penalty, and separate nonlinear radial transformations can remove a raw linear span relation. The empirical comparison therefore evaluates the complete feature transformation and fitted decoder as well as the reconstruction map.

5 Bounded residual forecasting

5.1 Separate bounded feature blocks

Each feature block is centered and scaled coordinatewise using A alone; feature standard deviations are floored at 10^{-6}. A tilde denotes this coordinatewise standardization. For a vector $v \in \mathbb{R}^p$, define

$$T_R(v) = \frac{v}{\sqrt{1 + \|v\|_2^2/R^2}}, \qquad R = \sqrt{p}. \tag{5}$$

This radial map has norm at most R and is nonexpansive with respect to its argument. If the common and departure block dimensions are p_0 and p_d, respectively, set $R_0 = \sqrt{p_0}$ and $R_d = \sqrt{p_d}$. The bounded feature vectors are

$$\psi_K(z) = \operatorname{concat}\big(T_{R_0}(\widetilde{\psi}_0(z)), T_{R_d}(\widetilde{d}_K(z))\big), \qquad \psi_R(z) = \operatorname{concat}\big(T_{R_0}(\widetilde{\psi}_0(z)), T_{R_d}(\widetilde{d}_R(z))\big).$$

The two blocks are transformed separately and are not standardized again. Consequently, appending a departure does not rescale the common features. Three partitions with 2, 4, and 8 local models, local rank two, and at most 24 references per local model give $p_0 = 42$ and $p_d = 168$ for the 12-dimensional histories in this experiment. The geometry is fitted without targets on A and remains fixed during selection and evaluation.

5.2 Fitting and calibrating the correction

For horizon h, let $f_{0,h}$ be the raw-history ridge baseline. Given an unstandardized history x, let $\psi(x)$ denote the chosen bounded feature vector after applying its A-fitted input scaler. The forecast is

$$\widehat{y}_{t+h} = f_{0,h}(x_t) + \gamma_h q_h(x_t), \qquad q_h(x) = b_h + \psi(x)^\top w_h. \tag{6}$$

Here $b_h \in \mathbb{R}$ is a free intercept, w_h is the coefficient vector, and γ_h controls the correction strength. For n_A fitting histories, the residual targets are $e_{ih} = y_{i,h} - f_{0,h}(x_i)$, where $y_{i,h}$ is the observed future at horizon h

for fitting history i. The correction minimizes

$$\frac{1}{n_A}\sum_{i=1}^{n_A}\{e_{ih} - b_h - \psi(x_i)^\top w_h\}^2 + \alpha_h\|w_h\|_2^2.$$

Set B selects $\alpha_h \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ by forecast error separately for each horizon. Set C then selects $\gamma_h \in \{0, 1/4, 1/2, 1\}$. Neither selection step refits the geometry or decoder, and set D is reserved for final evaluation.

5.3 A uniform bound including the intercept

Define the fitting residual mean, centered root mean square, and feature mean as

$$\bar{e}_h = \frac{1}{n_A}\sum_i e_{ih}, \qquad \sigma_{e,h} = \left[\frac{1}{n_A}\sum_i (e_{ih} - \bar{e}_h)^2\right]^{1/2}, \qquad \bar{\psi} = \frac{1}{n_A}\sum_i \psi(x_i).$$

Proposition 4 (Centered ridge correction bound)**.** *For the mean-squared-error objective above with $n_A \geq 1$, $\alpha_h > 0$, and an unpenalized intercept, suppose that the feature map satisfies $\|\psi(x)\|_2 \leq R$ on its input domain. For any scalar γ_h, the fitted coefficients and correction satisfy*

$$\|w_h\|_2 \leq \frac{\sigma_{e,h}}{2\sqrt{\alpha_h}}, \qquad |\gamma_h q_h(x)| \leq |\gamma_h|\left[|\bar{e}_h| + (R + \|\bar{\psi}\|_2)\frac{\sigma_{e,h}}{2\sqrt{\alpha_h}}\right]. \tag{7}$$

For a concatenated bounded common and departure vector, $R = \sqrt{R_0^2 + R_d^2}$; for the common-only vector, $R = R_0$. The free intercept contributes the residual-mean term and cannot be omitted. This bound controls the correction, not the baseline, total forecast error, or sensitivity of the preceding reconstruction to changes in the history.

Proof idea. Eliminating the intercept gives $b_h = \bar{e}_h - \bar{\psi}^\top w_h$. If Ψ_c is the matrix whose rows are $(\psi(x_i) - \bar{\psi})^\top$ and e_c is the centered residual vector, then $w_h = (\Psi_c^\top \Psi_c + n_A\alpha_h I)^{-1}\Psi_c^\top e_c$. The coefficient operator has singular values $s/(s^2 + n_A\alpha_h) \leq 1/(2\sqrt{n_A\alpha_h})$. This yields the coefficient bound. Applying Cauchy–Schwarz to $q_h(x) = \bar{e}_h + (\psi(x) - \bar{\psi})^\top w_h$ yields the correction bound. Appendix A gives the complete argument, including the nonexpansiveness of the radial map.

The theorem applies separately to each selected horizon and is a property of bounded-feature ridge regression. A summed-squared-error objective would need penalty $n_A\alpha_h$ to define the same estimator. The radial transformation is nonexpansive in its own input, but this does not give a Lipschitz bound for the preceding signed reconstruction or for its fallback boundary.

Baseline recovery and predictive gain. The calibration grid includes $\gamma_h = 0$, which recovers the same fitted baseline exactly. This is an available forecast rather than a guarantee that a finite calibration set selects the best future forecast. The coefficient and correction bounds likewise make no claim about prediction accuracy. The experiment therefore evaluates error, numerical coverage, and the correction inequalities as separate outcomes.

6 Matched forecasting evaluation

6.1 Comparison procedures

The primary candidate is the bounded KAHM departure. Its five primary comparators are the common features, bounded RBF departure, raw-history RBF forecasting, histogram gradient boosting, and a validation-selected non-KAHM procedure. Raw ridge, persistence, and KAHM and RBF variants that omit only the departure bound complete the ten-method comparison. The principal experiment uses the bounded features and fixed geometry defined in Sections 3 and 5. All non-ridge candidates are calibrated on C by blending their

Table 1: Data roles in each principal task suite. Each episode has 72 observations. A history contains 12 observations and predicts horizons 1, 4, and 12. The final evaluation set is not used to choose any model setting or correction weight.

Set	Episodes	Histories	Role
A	64	512	Fit scalers, reference geometry, baseline, and candidate decoders.
B	32	256	Choose hyperparameters separately for each forecast horizon.
C	32	256	Choose correction weights and one complete non-KAHM procedure.
D	128	1,664	Evaluate fixed forecasts and compute the reported losses.

forecasts with the same ridge baseline. The selection policy chooses one complete, calibrated non-KAHM procedure using horizon-averaged normalized C error; neither KAHM variant is eligible. Raw RBF and boosting have the same A/B/C roles and horizon-specific hyperparameter selection. Appendix B specifies their grids and costs.

6.2 Observation laws and independent evaluation roles

The experiment comprises four laws, two additional-noise levels, and 24 independent repetitions, giving 192 task suites. The nonlinear laws are an autoregression, Van der Pol dynamics, and Duffing dynamics. A stable linear Gaussian AR(2) process provides a negative control. Additional Gaussian observation noise has standard deviation ηs_A, where s_A is the standard deviation of all original A episode values and $\eta \in \{0, 0.1\}$. The added noise affects histories and targets. Underlying episodes, standardized noise draws, retained rows, and fitting seeds are paired between the two noise levels. The oscillator observations already contain sensor noise when $\eta = 0$.

Sets A, B, C, and D contain 64, 32, 32, and 128 episodes, respectively, each of length 72. Their base episodes and standardized noise draws use independent random streams; the additional-noise amplitude is estimated on A and shared across sets. Histories have length 12, horizons are $\mathcal{H} = \{1, 4, 12\}$, and consecutive forecast origins are four steps apart. A, B, and C retain at most 512, 256, and 256 histories, respectively; D retains all 1,664 origins. The protocol, executable source, analysis, and prepared inputs were recorded before reserved forecasts were computed. This was a prospective local specification informed by earlier studies, rather than public preregistration or external replication. Results from other studies are not pooled with this one.

6.3 Loss aggregation and simultaneous inference

For task τ, method m, and horizon h, let $\mathrm{MSE}_{D,\tau,m,h}$ be the mean squared error over D origins. Define $v_{\tau h} = \max\{\mathrm{Var}_{A,\tau}(Y_h), 10^{-12}\}$, where Y_h denotes the horizon-h target and the variance uses the retained A target rows with divisor n_A. The task loss is

$$L_{\tau m} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \frac{\mathrm{MSE}_{D,\tau,m,h}}{v_{\tau h}}.$$

For repetition r, let L^{NL}_{rm} average the six nonlinear law–noise task losses equally, and let L^{lin}_{rm} average the two linear-control task losses. For each of the five nonlinear comparators, define the useful-improvement contrast

$$d_{rm} = 0.97 L^{\mathrm{NL}}_{rm} - L^{\mathrm{NL}}_{rK}.$$

A positive population mean of this contrast corresponds to an improvement in expected normalized loss exceeding the specified 3% margin. The linear-control contrast is

$$d_{r,\mathrm{lin}} = 1.05 L^{\mathrm{lin}}_{r,\mathrm{ridge}} - L^{\mathrm{lin}}_{rK},$$

which tests noninferiority with a 5% relative tolerance. For each contrast, $\bar{d} = 24^{-1} \sum_r d_r$ and $s_d^2 = 23^{-1} \sum_r (d_r - \bar{d})^2$ denote the sample mean and variance. The one-sided lower confidence bound is

$$\bar{d} - t_{23,1-0.05/6} \frac{s_d}{\sqrt{24}},$$

Table 2: The six primary comparisons for bounded KAHM departure. L_K and L_m are losses averaged over repetitions. The first five rows use the nonlinear aggregate and $\bar{d} = 0.97L_m - L_K$; the last uses the linear aggregate and $\bar{d} = 1.05L_m - L_K$. Gain is $100(1 - L_K/L_m)$, so negative values favor the comparator. The lower bound concerns the absolute margin contrast, not the percentage gain. All bounds use 24 independent repetitions, 23 degrees of freedom, and a one-sided Bonferroni adjustment for six comparisons. A criterion is satisfied only when its lower bound is positive.

Comparator	L_K	L_m	Gain (%)	$\bar{d}$	Lower bound	Pass
Common basis	0.02993	0.03423	12.58	0.00328	0.00295	Yes
RBF departure (bounded)	0.02993	0.03028	1.17	-0.00055	-0.00079	No
Raw RBF	0.02993	0.03115	3.92	0.00029	-8.286×10^{-5}	No
Boosted trees	0.02993	0.02941	-1.77	-0.00140	-0.00194	No
C-selected non-KAHM policy	0.02993	0.02822	-6.04	-0.00255	-0.00299	No
Linear control: ridge	0.89810	0.89632	-0.20	0.04303	0.04161	Yes

where $t_{\nu,q}$ is quantile q of the Student t distribution with ν degrees of freedom. Bonferroni adjustment covers the six contrasts simultaneously at nominal level 95%, subject to the paired-mean t approximation. The sampling units are independent repetition aggregates; overlapping windows and paired noise levels do not increase the inferential sample size.

The prespecified criterion requires all six lower bounds to exceed zero. It also requires complete finite predictions, satisfaction of the feature, coefficient, and correction bounds, and a taskwise maximum absolute KAHM error no greater than five times the corresponding ridge maximum. The last requirement is an engineering diagnostic rather than a population risk guarantee. A wholly failed method or a final prediction-cell recovery prevents an efficacy conclusion. The 3% improvement margin, 5% tolerance, and fixed 24-repetition budget express the study's research criteria; they do not derive from a deployment loss function or a power calculation. Failure to satisfy the complete criterion concludes the planned performance evaluation without adding repetitions or selecting favorable conditions after inspecting D.

6.4 Predictive results

The bounded KAHM departure attains nonlinear aggregate normalized mean squared error 0.02993. Its point relative gains over the common basis, matched bounded RBF departure, raw RBF, boosted trees, and the C-selected non-KAHM procedure are 12.58%, 1.17%, 3.92%, −1.77%, and −6.04%, respectively. Positive gains denote smaller mean KAHM loss. These are point estimates; the simultaneous margin bounds are reported in Table 2. Of the five nonlinear improvement criteria, 1 is satisfied; the linear-control noninferiority criterion passes. All feature-, coefficient-, correction-, and maximum-error checks are satisfied. The results therefore do not satisfy the prespecified overall improvement criterion.

The improvement over the common basis shows that the bounded reconstruction departure provides useful functions to this decoder. The matched RBF reconstruction provides much of the same benefit: KAHM's point improvement over it is below the 3% margin. KAHM's point improvement over raw RBF exceeds 3%, but the simultaneous lower bound remains below zero, leaving that margin unresolved. Boosting and the selected non-KAHM procedure have smaller aggregate point losses. Thus the results support reconstruction features without establishing the required practical advantage specific to KAHM.

Figure 1 shows every law and noise level for the matched reconstruction control and the selected non-KAHM procedure. These descriptive comparisons show how the relative error varies across the declared conditions. The appendix gives all ten methods, and the accompanying data include every horizon-specific result.

6.5 Numerical checks and measured resources

The maximum taskwise KAHM-to-ridge maximum-error ratio is 1.553, below the limit of five. There are 0 feature-bound, 0 coefficient-bound, and 0 correction-bound violations. C selects a nonzero KAHM correction weight in 432 of 432 nonlinear task–horizon combinations and 54 of 144 linear-control combinations. This

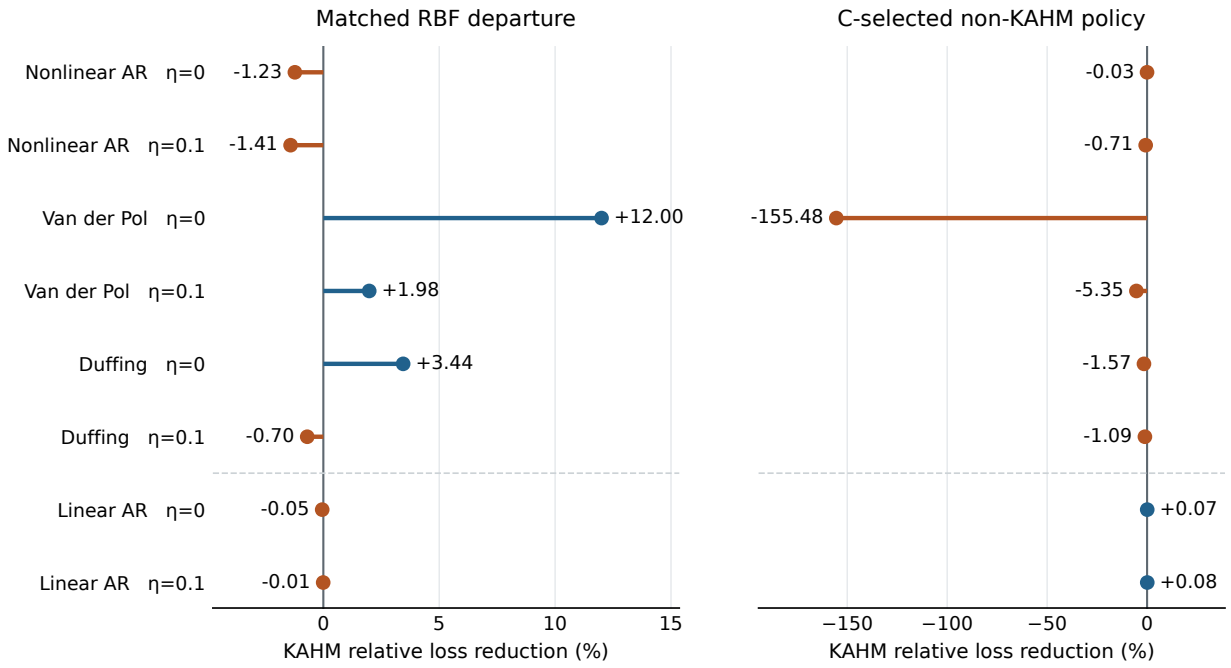


Figure 1: Condition-specific gains of bounded KAHM departure over the matched bounded RBF departure and the C-selected non-KAHM procedure. Each point is $100(1 - \overline{L}_K/\overline{L}_m)$, where the bars denote mean loss over 24 repetitions of that condition. Positive values favor KAHM. The panels use different horizontal scales, and labels give the point estimates. These descriptive estimates have no condition-specific confidence claims. The dashed divider separates nonlinear laws from the linear negative control.

activation frequency describes calibration choices, not a guarantee of lower D loss. Mean charged fitting and selection time is 0.0833 seconds for bounded KAHM departure and 0.0658 seconds for bounded RBF departure. Appendix B reports prediction time, retained model size, and the accounting boundaries.

The algebraic example, deterministic bound, and empirical comparisons answer different questions. A distinct feature can be poorly aligned with the target or difficult to estimate from the available sample. A bounded correction can remain inaccurate. An unresolved improvement margin does not establish equality of the representations. The conclusions therefore apply to the specified reconstructions, decoder, selection rules, and observation laws.

7 Supporting evidence and limits of transfer

Complementary studies examine the information retained by geometric features, access by finite decoders, transfer to natural series, and learning or updating the geometry. They use different representations, baselines, splits, loss scales, and sampling units from the bounded reconstruction-departure experiment. Their outcomes therefore describe separate fitted procedures; they are not pooled into its confidence family. Table 3 summarizes the evidence. The companion *Supplementary Studies* document provides the complete definitions, proofs, protocols, comparison tables, and failure accounting.

Information and finite-decoder access. In a controlled branch construction, the same observations and noise support targets that vary from branch identity to signed position. An affine reconstruction can be nearly exact while its discrepancy-based code loses the position required for prediction. Affine local coordinates recover accurate prediction in that example, so the benefit is not exclusive to nonlinear reconstruction. In a separate finite-decoder comparison, KAHM local features outperform raw and affine local features on Van der Pol at both 64 and 256 Gaussian landmarks. Raw histories perform better on nonlinear autoregression,

Table 3: Scope and outcomes of the supporting studies. Counts describe the corresponding design, not independent forecasting domains. Natural source aggregates and post-hoc sensitivities are descriptive.

Study	Evaluation scope	Interpretation
Information and decoder capacity	Target interventions and 24 full repetitions per forecasting setting	Retaining local position can help; KAHM gains depend on the law and decoder.
Natural transfer and scaling	Eight source families; four reserved for confirmation; 208 configurations	Raw plus KAHM has higher reserved-source loss than raw RBF. A post-hoc scaling change preserves that ordering.
Additive correction	144 synthetic and 24 archive configurations	Some synthetic gains; two incomplete archive configurations prevent the planned full-roster comparison.
Proxy metric and donor residuals	48 tasks and 16 procedures	Corrections improve ridge, but incremental benefits of the learned metric and donor residuals are unresolved.
Direct reconstruction training	46 planned tasks; 43 feasible	Training helps KAHM in some settings; equally trained local controls remain competitive or better.
Online adaptation	Twelve streams, six methods, two capacities	Both capacity settings fail their accuracy criteria; extreme extrapolation persists after a solver correction.

and both controls perform better on Duffing. The inferential units are 24 independent full repetitions per setting, with separate multiplicity adjustments for the original and follow-up studies.

Natural-data transfer and model selection. The reserved-source comparison uses Traffic, Hospital, Saugeen River flow, and US births. Raw-history augmentation with KAHM features has 4.26% higher source-balanced scaled MSE than raw RBF and costs 2.30 times as much in the measured warm-library workflow. Raw ridge is more accurate than both. The favorable development mean is strongly influenced by FRED-MD, and it does not establish a KAHM-specific benefit. Adding geometric descriptors to metadata and a paid validation probe changes none of the 80 confirmation decisions. The post-hoc min–max scaling analysis reuses these sources and simulation draws; it changes individual errors but does not reverse the principal mean-loss orderings.

Residual correction and learned geometry. An additive correction preserves the fitted raw baseline as a candidate, but validation can select a correction that worsens test error. The synthetic factorial resolves gains over raw RBF in five of twelve comparisons, while a simpler geometric correction has lower mean error in every Van der Pol and Duffing cell. On the archive roster, the two capacities of one Car Parts series produce eight failed geometric fits and four unavailable policy outputs. The planned full-roster geometric and policy losses are consequently undefined. A post-hoc comparison on 22 successful configurations gives additive KAHM only a 0.31% gain over raw RBF at 13.29 times its measured model-pipeline cost; ridge is more accurate and cheaper on that subset.

The donor-residual study completes all 48 tasks. Its learned-metric, out-of-fold KAHM procedure improves the ridge reference on the three synthetic laws and has 2.18% lower descriptive archive loss. However, the metric-versus-identity and donor-versus-in-sample intervals include zero. No adjusted comparison with an equally trained geometric control favors KAHM. These results support residual correction in the evaluated settings without identifying the proposed metric or KAHM reconstruction as the necessary source of that improvement.

Training and adaptation. Direct reconstruction training produces two favorable, six unfavorable, and ten unresolved primary contrasts. Neither favorable contrast is against an equally shape-trained non-KAHM local control. Three selected rideshare series have no eligible training-query windows, leaving the planned four-collection natural aggregate undefined; results on the seven available series form a separate descriptive

comparison. A nine-task gate-training pilot uses a separate generator and budget and does not contribute to those intervals.

The online study meets its declared 99th-percentile service-time screen, but both capacity settings fail the recurring-stream and natural-data accuracy criteria. A corrected ridge solver removes computational exceptions while retaining an extreme forecast that a fresh solve also reproduces. Thus a nonvanishing reconstruction denominator and an accurate linear solve do not by themselves control feature extrapolation. The bounded, fixed-geometry decoder analyzed in the main experiment is a different procedure; its correction bound does not validate a repair of the online method.

8 Discussion and conclusion

The central results distinguish three properties of a reconstruction representation. Algebraic distinctness means that its coordinates permit functions unavailable to a specified comparison span. A bounded correction means that a fitted residual head cannot exceed a stated magnitude under the feature-norm assumption. Predictive usefulness means that the complete procedure achieves lower loss under a specified evaluation distribution. None of these properties implies the next one without additional assumptions or evidence.

The principal experiment supports a contribution from reconstruction departures beyond the shared locality and principal-coordinate features. It does not establish the prespecified practical advantage specific to KAHM across the comparison family. The matched RBF reconstruction achieves much of the same benefit, and both boosting and the selected non-KAHM procedure have lower aggregate point loss. Failure to establish a margin does not establish equality of two methods. Likewise, the irregular-bank example demonstrates existence of distinct raw spans, not distinctness for every bank fitted in the experiment.

The correction bound provides a precise magnitude guarantee for the specified decoder. It does not establish robustness of the full history-to-feature map: signed normalization and the affine fallback may still be sensitive to the input. The baseline and targets are not uniformly bounded, so the theorem gives no uniform forecast-error bound. Its applicability to other bounded features also limits a claim of methodological novelty based on the bound alone.

The empirical scope is finite. Twenty-four independent repetitions quantify variability under four fixed simulation laws; they are not 24 independent application domains. The observation maps, finite burn-in, history length, horizons, model grids, and resource settings are part of the tested procedures. The simultaneous intervals use a paired-mean t approximation, and the 3% improvement and 5% noninferiority margins express research criteria rather than application-specific requirements. The supporting natural-data and adaptive studies impose further limits on extrapolation of the simulated gains. The experiments are not a comprehensive comparison with neural, pretrained, long-context multivariate, or modern adaptive forecasting methods.

Computation must also be interpreted at the level measured. Fitting costs include the candidates evaluated during selection. Prediction time, serialized state size, owned array storage, and whole-process peak memory measure different resources and cannot be substituted for one another. The measurements characterize the supplied implementations on the recorded environment; they do not establish minimum algorithmic costs or satisfy an unspecified deployment budget.

KAHM reconstruction can therefore supply useful features to a finite forecasting decoder, with a value that depends on geometry, feature parameterization, regularization, and target structure. The contribution of the present study is to characterize these dependencies and evaluate them against matched alternatives. A general forecasting advantage specific to KAHM remains unestablished by the reported evidence.

Reproducibility and research transparency

The anonymous saved-forecast supplement contains predictions and evaluation targets for all 192 task suites, along with fitting targets that determine the loss-normalization scales. Its standalone program, `verify_saved_forecasts.py`, reconstructs eight numerical analysis tables and all six primary contrasts, verifies array identities, and checks that each policy forecast equals its recorded selected procedure. The

supplement includes all ten procedures and all declared conditions, regardless of predictive performance. The companion *Supplementary Studies* provides the complete methods and results for the supporting investigations summarized in Section 7.

The principal study's locally recorded source and input inventory contains 239 files. The saved evidence contains 1,920 prediction arrays and 9,584,640 forecast values. The recorded selected-state replay and separately implemented calculations provide computational consistency checks. Reconstructing scores from saved forecasts, replaying selected states, and refitting all model candidates are distinct operations. The anonymous numeric supplement performs the first; it does not refit models or reproduce feature-bound calculations and resource timings. The complete fitting specification and numerical conventions are given in Appendix B. The prospective record was local, informed by the supporting studies, and not publicly preregistered.

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A Reconstruction algebra and bounded residual decoding

This appendix analyzes fixed reference maps and their use by a specified decoder. It establishes normalized-kernel identities, conditions for equality or inequality of feature spans, and a bound on fitted residual corrections. None of these statements assumes a particular time-series distribution or guarantees a forecasting improvement. Unless another subscript is displayed, vector norms are Euclidean norms. The numerical examples can be evaluated with `extensions/reconstruction_mechanism/algebra.py`; their reference banks are constructed independently of forecasting outcomes.

A.1 Normalized kernel identities

Let $q_1, \dots, q_M \in \mathbb{R}^d$ be reference histories, and let $Q \in \mathbb{R}^{M\times d}$ contain their transposes as rows. Let $P \in \mathbb{R}^{d\times r}$ have orthonormal columns, with $1 \le r \le d$, and let $W \in \mathbb{R}^{r\times r}$ be a symmetric positive-definite precision matrix for the projected coordinates. For a column-vector query $z \in \mathbb{R}^d$, define the Gaussian kernel vector $k(z) \in \mathbb{R}^M$ by

$$k_i(z) = \exp\left[-\frac{(P^\top(z-q_i))^\top W(P^\top(z-q_i))}{2r}\right], \qquad i = 1, \dots, M.$$

Let $K \in \mathbb{R}^{M\times M}$ be the reference Gram matrix, with $K_{ij} = k_i(q_j)$. For $\lambda > 0$, set $C = (K + \lambda I_M)^{-1}$, where I_M is the $M \times M$ identity matrix, and let $\mathbf{1}_M$ be the length-M vector of ones. The matrix K is symmetric positive semidefinite, so C exists and is symmetric positive definite. The KAHM and matched positive-kernel reconstructions are

$$A_K(z) = \frac{Q^\top C k(z)}{\mathbf{1}_M^\top C k(z)}, \qquad A_R(z) = \frac{Q^\top k(z)}{\mathbf{1}_M^\top k(z)}. \tag{8}$$

The first expression is defined wherever its signed denominator is nonzero. The second denominator is positive for every finite query. The two maps share the references and projected Gaussian kernel. Multiplying $k(z)$ by a common positive scalar cancels in either ratio, which justifies subtracting the largest log-kernel value before query evaluation.

Target-transfer identity. For a query satisfying $s(z) = \mathbf{1}_M^\top Ck(z) \neq 0$, define the normalized weight vector $\omega(z) = Ck(z)/s(z)$. Its entries sum to one. For any reference target vector $y \in \mathbb{R}^M$,

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I_M)^{-1} y}{k(z)^\top (K + \lambda I_M)^{-1} \mathbf{1}_M}.$$

Thus target transfer is the ratio of two dual kernel-ridge predictions with regularization matrix $K + \lambda I_M$ and no additional intercept: one fitted to y, and one fitted to the constant target $\mathbf{1}_M$. For an affine scalar function $f(q) = q^\top b + c$, with $b \in \mathbb{R}^d$ and $c \in \mathbb{R}$, let $f(Q) = Qb + c\mathbf{1}_M$ be its reference values. Then

$$\omega(z)^\top f(Q) - f(z) = \{A_K(z) - z\}^\top b.$$

Proof of Proposition 1. The definition gives $\mathbf{1}_M^\top \omega(z) = s(z)/s(z) = 1$. Because $C = C^\top$, $\omega(z)^\top y = k(z)^\top Cy/s(z)$ and $s(z) = k(z)^\top C\mathbf{1}_M$, proving the ratio identity. For the affine function,

$$\omega(z)^\top f(Q) = \omega(z)^\top Qb + c\omega(z)^\top \mathbf{1}_M = A_K(z)^\top b + c.$$

Subtracting $z^\top b + c$ proves the final identity. □

This identifies direct target transfer as a normalized kernel-ridge construction. The ratio generally differs from the unnormalized kernel-ridge predictor because its denominator depends on the query. Similarly, $A_K(z)^\top b$ is normalized kernel ridge applied to the restricted reference target Qb. For any reconstruction A, the identity $A(z) + \{z - A(z)\} = z$ also implies a linear dependence among the raw input, reconstruction, and residual when all three are included as unrestricted linear features. These identities constrain claims about the resulting function class while allowing useful parameterizations for a restricted or regularized decoder.

A.2 When reconstruction departures have the same span

Let $\mu \in \mathbb{R}^d$ be the reference center and define the affine reconstruction $A_A(z) = \mu + PP^\top(z - \mu)$. Fix a common scalar locality gate $g(z)$, and define column-vector departures

$$d_K(z) = g(z)\{A_K(z) - A_A(z)\}, \qquad d_R(z) = g(z)\{A_R(z) - A_A(z)\}.$$

A feature span means the set of scalar functions obtained by finite linear combinations of its coordinate functions. Let $\mathcal{S}_0$ be the span of the gate $g(z)$ and the coordinates of $g(z)P^\top(z - \mu)$. It contains every coordinate of $g(z)A_A(z)$.

Equal augmented feature spans. Suppose $K\mathbf{1}_M = \kappa\mathbf{1}_M$ for a scalar κ and C leaves the augmented reference-coordinate space $\mathcal{V} = \mathrm{span}\{\mathbf{1}_M, Q_{:1}, \ldots, Q_{:d}\} \subseteq \mathbb{R}^M$ invariant. Here $Q_{:j}$ denotes column j of Q. Then there exist $B \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$ such that

$$C\mathbf{1}_M = a\mathbf{1}_M, \qquad CQ = QB + \mathbf{1}_M c^\top, \qquad a = (\kappa + \lambda)^{-1} > 0.$$

For these coefficients,

$$A_K = \frac{B^\top A_R + c}{a}, \qquad d_K = \frac{B^\top}{a} d_R + \left(\frac{B^\top}{a} - I_d\right) gA_A + \frac{gc}{a}.$$

The span obtained by adjoining the coordinates of d_K to $\mathcal{S}_0$ is the same as that obtained by adjoining the coordinates of d_R.

Proof of Proposition 2. The constant-row-sum assumption implies $(K+\lambda I_M)\mathbf{1}_M = (\kappa+\lambda)\mathbf{1}_M$, so $C\mathbf{1}_M = a\mathbf{1}_M$. Since K is positive semidefinite, $\kappa \geq 0$ and hence $a > 0$. Invariance of $\mathcal{V}$ means each column of CQ can be expressed using Q and $\mathbf{1}_M$, which gives B and c. Using symmetry of C in Equation (8),

$$A_K(z) = \frac{(CQ)^\top k(z)}{(C\mathbf{1}_M)^\top k(z)} = \frac{B^\top Q^\top k(z) + c\mathbf{1}_M^\top k(z)}{a\mathbf{1}_M^\top k(z)} = \frac{B^\top A_R(z) + c}{a}.$$

Subtract A_A and multiply by g to obtain the displayed departure identity. Since $gA_A = g\mu + P\{gP^\top(z-\mu)\}$, all its coordinates belong to $\mathcal{S}_0$. Thus the KAHM-augmented span is contained in the RBF-augmented span.

For the reverse inclusion, the restriction of the invertible matrix C to the invariant finite-dimensional space $\mathcal{V}$ is injective and therefore surjective onto $\mathcal{V}$. Its images $C\mathbf{1}_M = a\mathbf{1}_M$ and the columns of CQ consequently span $\mathcal{V}$. Hence there exist $D \in \mathbb{R}^{d\times d}$ and $e \in \mathbb{R}^d$ such that $Q = (CQ)D + \mathbf{1}_M e^\top$. Multiplying its transpose by $k(z)/(\mathbf{1}_M^\top k(z))$ gives $A_R(z) = aD^\top A_K(z) + e$. After multiplication by g and subtraction of gA_A, this expresses each RBF departure coordinate using KAHM departure coordinates and $\mathcal{S}_0$. The spans therefore agree, including when the original reference-coordinate columns are redundant. □

For example, take scalar references $q = (-1, 1)^\top$, $P = 1$, $g = 1$, and $A_A(z) = z$. If the off-diagonal Gram entry is ρ, then $K\mathbf{1}_2 = (1+\rho)\mathbf{1}_2$ and $Kq = (1-\rho)q$. Therefore

$$A_K = c_\rho A_R, \qquad d_K = c_\rho d_R + (c_\rho - 1)z, \qquad c_\rho = \frac{1+\rho+\lambda}{1-\rho+\lambda}. \tag{9}$$

The coefficient c_ρ is positive, so $\operatorname{span}(1, z, d_K) = \operatorname{span}(1, z, d_R)$. With the sample-variance precision $1/2$, $\rho = e^{-1}$. The regularization iteration used by the implementation yields $\lambda = 2.6519665429866075$ and $c_\rho = 1.224037566462905$. The maximum numerical identity error on the constructed query grid is below 6×10^{-16}.

These are equalities between untransformed feature spans. An isotropic ridge penalty is not invariant under a general nonorthogonal change of coordinates. Applying separate nonlinear radial transformations can also remove the displayed linear relation. A forecasting comparison therefore evaluates the complete reconstruction, feature transformation, and fitting procedure, even when two untransformed feature sets have the same span.

A.3 An explicit distinct reconstruction basis

Irregular scalar references. Let $q_1 < q_2 < q_3$ with $q_1 + q_3 \neq 2q_2$, and define $k_i(z) = \exp\{-t(z-q_i)^2/2\}$ for precision $t > 0$. Let $K_{ij} = \exp\{-t(q_i - q_j)^2/2\}$, $C = (K + \lambda I_3)^{-1}$ for $\lambda > 0$, and suppose every entry of $v = C\mathbf{1}_3$ is positive. For the reconstructions in Equation (8), with $Q = q = (q_1, q_2, q_3)^\top$, gate $g = 1$, and $A_A(z) = z$,

$$A_K \notin \operatorname{span}(1, z, A_R), \qquad d_K \notin \operatorname{span}(1, z, d_R).$$

These are statements about scalar functions on the whole real line.

Proof of Proposition 3. Set $u = Cq$. Since C is symmetric, $A_K(z) = (k(z)^\top u)/(k(z)^\top v)$, and positivity of v makes its denominator positive for every z. The six sums $q_i + q_j$ with $i \leq j$ are distinct: strict ordering of the three references orders all the sums except possibly $q_1 + q_3$ and $2q_2$, and their equality is excluded by assumption. Products of Gaussian translates satisfy

$$k_i(z)k_j(z) = e^{-tx^2} e^{-t(q_i^2+q_j^2)/2} e^{t(q_i+q_j)z}.$$

After removal of the common positive factor e^{-tx^2}, these products are nonzero multiples of exponentials with six distinct real exponents. Such exponentials are linearly independent. Indeed, differentiating a vanishing linear combination zero through fifth order at $z = 0$ gives a Vandermonde system whose determinant is the product of the nonzero pairwise exponent differences.

Suppose first that $A_K = cA_R + d$ for scalars c, d. Define $w = u - dv$. Multiplication by the two positive denominators gives

$$\left(\sum_i k_i w_i\right)\left(\sum_j k_j\right) = c\left(\sum_i k_i q_i\right)\left(\sum_j k_j v_j\right).$$

Independence of the six products implies equality of their coefficients. The diagonal coefficients give $w_i = cq_i v_i$, and the off-diagonal coefficients give $w_i + w_j = c(q_i v_j + q_j v_i)$ for $i < j$. If $c = 0$, all w_i vanish, so $Cq = dC\mathbf{1}_3$. Invertibility of C would imply $q = d\mathbf{1}_3$, contrary to the distinct references. If $c \neq 0$, substituting the

diagonal equations into the off-diagonal ones yields $(q_i - q_j)(v_i - v_j) = 0$. Thus $v = a\mathbf{1}_3$ for some $a > 0$, and $(K + \lambda I_3)v = \mathbf{1}_3$ implies that K has constant row sums. This is impossible: its middle-row sum exceeds the first-row sum by $K_{23} - K_{13} > 0$ and the last-row sum by $K_{12} - K_{13} > 0$, since the Gaussian decreases strictly with positive distance. This rules out every affine relation $A_K = cA_R + d$.

Both reconstructions are bounded functions of z. The RBF reconstruction is a convex combination of q_1, q_2, q_3. Also,

$$A_K(z) = \frac{\sum_i k_i(z) v_i \, (u_i/v_i)}{\sum_i k_i(z) v_i}$$

is a convex combination of the three finite numbers u_i/v_i. If $A_K = cA_R + d + bx$ held on the real line, boundedness of both reconstructions would force $b = 0$. The affine relation already ruled out would then follow. This proves the first span statement. Finally, $d_K = A_K - z$ and $d_R = A_R - z$ imply $\operatorname{span}(1, z, d_R) = \operatorname{span}(1, z, A_R)$. Membership of d_K in this span would imply membership of A_K, a contradiction. □

A concrete example uses $q = (-1, 0, 2)^\top$, sample-variance precision $t = 3/7$, and $\lambda = 4.498274743339644$ from the specified regularization iteration. Positivity of $C\mathbf{1}_3$ can be verified without examining a query grid. Write $K = I_3 + E$ and $a = 1 + \lambda$. For a matrix, let $\|\cdot\|_\infty$ denote the maximum absolute row sum; for a vector, it denotes the largest absolute entry. If $a > 2\|E\|_\infty$, then the Neumann series for $aC = (I_3 + E/a)^{-1}$ gives

$$\|aC\mathbf{1}_3 - \mathbf{1}_3\|_\infty \le \sum_{j=1}^{\infty} (\|E\|_\infty/a)^j = \frac{\|E\|_\infty/a}{1 - \|E\|_\infty/a} < 1.$$

Every entry of $aC\mathbf{1}_3$ is then strictly positive. In the example, $\|E\|_\infty = e^{-3/14} + e^{-6/7} = 1.2314905926823394$, so the condition holds. On 81 constructed queries in $[-3, 4]$, the maximum absolute residual after least-squares projection of the sampled d_K values onto the sampled $(1, z, d_R)$ columns is 0.05355670997078987. This numerical illustration supplements the analytic existence result. It does not establish distinctness for every higher-dimensional bank used in the forecasting experiments.

A.4 A bounded correction with a free intercept

For a feature block $v \in \mathbb{R}^{p_b}$ of positive width p_b, define

$$T_{R_b}(v) = \frac{v}{\sqrt{1 + \|v\|_2^2/R_b^2}}, \qquad R_b = \sqrt{p_b}. \tag{10}$$

Its norm is at most R_b. To see its sensitivity to its own input, set $a = 1 + \|v\|_2^2/R_b^2$. Its Jacobian is $a^{-1/2} I - a^{-3/2} vv^\top/R_b^2$. For $v \neq 0$, the eigenvalue in the direction of v is $a^{-3/2}$, and each orthogonal direction has eigenvalue $a^{-1/2}$; at zero, the Jacobian is the identity. All its singular values are therefore at most one. Integrating the derivative along the segment between any two inputs proves $\|T_{R_b}(v) - T_{R_b}(v')\|_2 \le \|v - v'\|_2$. Thus the radial transformation is nonexpansive.

Concatenating separately transformed blocks gives a feature vector $\psi(z) \in \mathbb{R}^p$, where $p = \sum_b p_b$, satisfying $\|\psi(z)\|_2 \le R = (\sum_b R_b^2)^{1/2}$. This norm bound holds wherever the preceding feature construction is defined. Nonexpansiveness of the radial transformation does not by itself give a Lipschitz constant for the full history-to-feature map, which includes the signed reconstruction denominator and its fallback rule.

Centered ridge correction. For $n \ge 1$, let $e_i \in \mathbb{R}$ be residual targets and $\psi_i \in \mathbb{R}^p$ be training feature vectors. Fit $b \in \mathbb{R}$ and $w \in \mathbb{R}^p$ by minimizing

$$\frac{1}{n} \sum_{i=1}^{n} (e_i - b - \psi_i^\top w)^2 + \alpha \|w\|_2^2, \qquad \alpha > 0,$$

with no penalty on b. Define the training means $\bar{e} = n^{-1} \sum_i e_i$ and $\bar{\psi} = n^{-1} \sum_i \psi_i$, the residual vector $e = (e_1, \ldots, e_n)^\top$, its centered version $e_c = e - \bar{e}\mathbf{1}_n$, and its centered root mean square $\sigma_e = \|e_c\|_2/\sqrt{n}$. Suppose query features satisfy $\|\psi(z)\|_2 \le R$. Then, for any scalar γ,

$$\|w\|_2 \le \frac{\sigma_e}{2\sqrt{\alpha}}, \qquad |\gamma q(z)| \le |\gamma| \left[|\bar{e}| + (R + \|\bar{\psi}\|_2) \frac{\sigma_e}{2\sqrt{\alpha}} \right], \quad q(z) = \bar{e} + (\psi(z) - \bar{\psi})^\top w. \tag{11}$$

Proof of Proposition 4. For fixed w, differentiating with respect to the intercept gives $b = \bar{e} - \bar{\psi}^\top w$. Let $\Psi_c \in \mathbb{R}^{n\times p}$ have row i equal to $(\psi_i - \bar{\psi})^\top$. Substituting the fitted intercept reduces the objective to $n^{-1}\|e_c - \Psi_c w\|_2^2 + \alpha\|w\|_2^2$. Its unique minimizer is

$$w = (\Psi_c^\top \Psi_c + n\alpha I_p)^{-1}\Psi_c^\top e_c.$$

For a singular value $s \geq 0$ of Ψ_c, the corresponding singular value of the coefficient operator is $s/(s^2 + n\alpha)$. The inequality $(s - \sqrt{n\alpha})^2 \geq 0$ implies

$$\frac{s}{s^2 + n\alpha} \leq \frac{1}{2\sqrt{n\alpha}}.$$

The operator norm is therefore at most $1/(2\sqrt{n\alpha})$, giving $\|w\|_2 \leq \|e_c\|_2/(2\sqrt{n\alpha}) = \sigma_e/(2\sqrt{\alpha})$. Finally, the fitted intercept gives $q(z) = \bar{e} + (\psi(z) - \bar{\psi})^\top w$. Cauchy–Schwarz and the triangle inequality yield

$$|q(z)| \leq |\bar{e}| + (\|\psi(z)\|_2 + \|\bar{\psi}\|_2)\|w\|_2 \leq |\bar{e}| + (R + \|\bar{\psi}\|_2)\frac{\sigma_e}{2\sqrt{\alpha}}.$$

Multiplication by $|\gamma|$ proves the correction bound. □

The mean-squared-error objective corresponds to a summed-squared-error objective with penalty $n\alpha$. Keeping the numerical value α unchanged when switching to a summed loss would define a different estimator. For multiple forecast horizons, the bound applies separately to each fitted residual vector and each selected pair (α_h, γ_h). This is a general property of ridge regression on bounded features, including the KAHM features considered here. It bounds the additive correction but does not bound the retained raw baseline or total forecast error. The experiment checks the feature, coefficient, and correction inequalities on every saved task and reports forecast errors separately.

B Complete reconstruction-departure evaluation

This appendix specifies the observation laws, fitting procedure, complete comparison set, stability checks, and resource measurements for Section 6. The reproduction materials contain the implementation, saved predictions and model states, and machine-readable analyses under `extensions/reconstruction_mechanism`. All tables and figures in this appendix are generated from those analyses.

B.1 Observation laws and data roles

The linear negative control is

$$y_t = 0.65y_{t-1} - 0.15y_{t-2} + \epsilon_t, \qquad \epsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.1^2),$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2. The process starts with two zeros; 256 stored values are discarded, and 72 values beginning at index 256 are retained. The nonlinear autoregression is

$$y_t = \tanh\{1.25y_{t-1} - 0.65y_{t-2} + 0.25\sin(3y_{t-1}) + u_t\}, \qquad u_t \overset{\text{iid}}{\sim} U[-0.1, 0.1],$$

where $U[a, b]$ denotes the uniform distribution on $[a, b]$. Its two initial values are independent draws from $U[-0.8, 0.8]$, and 80 stored values are discarded before retaining an episode.

The Van der Pol dynamics are $\dot{x} = v$ and $\dot{v} = (1 - x^2)v - x$. The Duffing dynamics are $\dot{x} = v$ and $\dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi)$. Here x is position, v is velocity, a dot denotes differentiation with respect to continuous time t, and the Duffing phase φ is uniform on $[0, 2\pi]$. Initial position and velocity are independent draws from $U[-2, 2]$. The ordinary differential equations are integrated in float64 arithmetic using the classical fourth-order Runge–Kutta method with step 0.025, observation interval 0.1, and a discarded initial time interval of length ten. The scalar observation is $0.95\tanh(x) + \xi$, with independent sensor noise $\xi \sim U[-0.02, 0.02]$. The resulting numerical integration and observation map define the tested simulation law; the observations are not exact continuous-time trajectories.

For each law and repetition, the episode sets A, B, C, and D are generated once. Additional noise has the form $\eta s_A Z$, where $\eta \in \{0, 0.1\}$, $Z \sim \mathcal{N}(0,1)$ independently at each stored observation, and s_A is the population standard deviation of all A episode values before this addition. The underlying episodes and Z arrays are paired across the two noise levels. Added noise affects both histories and observed future targets. At $\eta = 0$, the oscillator sensor noise is still present. The B/C/D linear observations are Gaussian conditional on the A-estimated noise amplitude, while their marginal law across repetitions is a variance mixture. Finite burn-in does not imply exact stationarity, and the added Gaussian noise means that the observed responses are not uniformly bounded.

The 24 repetition seeds are 946100–946123; the development seed 975001 is excluded. Base episodes and standardized noise arrays for different sets use independent random streams. The additional-noise amplitude is estimated on A and shared across sets. The fitting seed and row selection are paired across noise levels. Using zero-based episode indices, the 13 origins $11, 15, \ldots, 59$ each have a length-12 history ending at the origin and targets at offsets 1, 4, and 12. A/B/C/D contain 64/32/32/128 episodes. The first three sets retain 512/256/256 histories; D retains all 1,664 histories. This gives 958,464 scored target values per method over the full experiment. Inference nevertheless uses 24 independent repetition aggregates.

A fits every scaler, reference bank, kernel factor, baseline, and decoder. B selects model settings separately by horizon. C selects each candidate's mixing coefficients and then one complete non-KAHM procedure. Fitted states are saved before D histories are evaluated, and D targets are used only after predictions are fixed. No model or selection rule is updated on D.

B.2 Reference geometry and numerical evaluation

The geometric bank first standardizes each history coordinate using its A mean and standard deviation. A standard deviation at most 10^{-12} is replaced by one. At most 512 A rows enter the bank fit. Separate k-means partitions request 2, 4, and 8 groups, using five initializations and at most 150 iterations. If there are too few distinct input rows, the requested group counts are reduced to the available number. An unresolved empty-group configuration is retried with halved group counts. Within each group, at most 24 reference rows are sampled without replacement. The references are stored in float32 and then used in float64 arithmetic. Each local model retains at most two principal directions of positive empirical variance. A group with no retained direction contributes an affine fallback and a zero departure.

For the row-stored references Q_j, define $\mu_j = n_j^{-1} \sum_{i=1}^{n_j} q_{ji}$ and

$$s_j = \max\left\{ \frac{1}{n_j} \sum_{i=1}^{n_j} \|q_{ji} - \mu_j\|_2^2, 10^{-6} \right\}.$$

This is the scale used in the common locality weights. The projected sample covariance is

$$S_j = \frac{1}{n_j - 1} \sum_{i=1}^{n_j} P_j^\top (q_{ji} - \mu_j)(q_{ji} - \mu_j)^\top P_j.$$

Its inverse defines the shared Gaussian kernel in Section 3. Kernels and references are identical between KAHM and positive-RBF reconstruction.

The KAHM regularizer is fitted from reference reconstruction, without forecast targets. Suppress the local index j, let $Q \in \mathbb{R}^{n \times d}$ contain the local references, and define the reference energy $\nu = \|Q\|_F^2/(nd)$, where $\|\cdot\|_F$ is the square root of the sum of squared matrix entries. Initialize $\lambda^{(0)} = 2.5\nu$. At iteration a, compute

$$C^{(a)} = (K + \lambda^{(a)} I_n)^{-1}, \qquad \lambda^{(a+1)} = 2\nu + \frac{\|Q - KC^{(a)}Q\|_F^2}{nd}.$$

The iteration stops when $|\lambda^{(a+1)} - \lambda^{(a)}| \le 10^{-3}$ or after 100 inverse computations. The reconstruction uses the last computed $C^{(a)}$ and its corresponding $\lambda^{(a)}$, rather than the subsequent proposal $\lambda^{(a+1)}$. Nonfinite factors or a nonpositive regularizer invoke the recorded local fallback. This rule determines a reference-based regularizer; it does not optimize forecasting loss.

For each query, the maximum log-kernel entry is subtracted before exponentiation. The resulting common positive rescaling leaves both normalized reconstruction ratios unchanged. Signed cancellation is checked using ρ_j in Equation (3). The affine fallback makes the corresponding raw departure zero if the ratio cannot be evaluated reliably. The experimental results below report how often this occurs.

B.3 Decoders, comparison grids, and descriptive losses

The common and departure feature columns are standardized separately using A, with each standard deviation replaced by its maximum with 10^{-6}. The blocks are then radially bounded as specified in Equation (5), with no subsequent scaling. The two unbounded-departure variants retain the bounded common block and omit only the departure's radial transformation.

Every ridge head uses mean squared error plus $\alpha\|w\|_2^2$ and an unpenalized intercept, with $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. Equivalently, the unnormalized sum-of-squares objective uses penalty $n\alpha$ for n fitting rows. Raw-history ridge and RBF methods standardize their input coordinates on A with a 10^{-6} scale floor. The raw RBF method concatenates standardized histories with at most 64 Nyström features, then standardizes the concatenated columns using A. The squared Gaussian bandwidth equals the median of all strictly positive squared landmark distances multiplied by 0.5, 1, or 2, with no bandwidth floor. When there are no strictly positive landmark distances, the median used by this rule is set to one. B selects this factor and the ridge penalty separately for each horizon. For a landmark Gram matrix whose largest eigenvalue is $\zeta_{\max}$, the Nyström transform retains eigenvalues above $\max\{10^{-12}, 10^{-10}\zeta_{\max}\}$. Let K_{MM} be the symmetrized landmark Gram matrix. Let the columns of V be its retained orthonormal eigenvectors, and let the diagonal matrix Λ contain their positive eigenvalues. If $k_M(u)$ is the vector of Gaussian similarities between a standardized query u and the M landmarks, its Nyström coordinates are $\Lambda^{-1/2}V^\top k_M(u)$. The Gaussian is $\exp\{-\|u-u'\|_2^2/(2\ell^2)\}$, with squared bandwidth ℓ^2 chosen by the median-distance rule above. These coordinates are concatenated with the standardized raw history before column scaling and ridge fitting.

Histogram gradient boosting uses 100 iterations and L2 regularization one. B selects learning rate 0.05 or 0.1 and maximum leaf count 7 or 15 separately by horizon. Persistence predicts the latest observed history value at every horizon. Every non-ridge candidate is then blended with ridge using a C-selected coefficient from $\{0, 1/4, 1/2, 1\}$. Ties are resolved by the recorded candidate order. The non-KAHM policy chooses among ridge, raw RBF, boosting, persistence, the common basis, and the two RBF-departure variants; both KAHM variants are excluded. Table 4 reports all methods and conditions.

Table 4: All ten procedures, every declared law and both added-noise levels. Entries are mean horizon-averaged A-variance-normalized D MSE over 24 repetitions; lower is better. The first six columns enter the nonlinear aggregate with equal weight. The final two form the separate linear control. These conditionwise summaries are descriptive. "Unbounded" omits only the departure radial bound. Persistence and raw nonlinear controls include their C-selected blend with ridge.

Procedure	Nonlinear AR		Van der Pol		Duffing		Linear AR	
Added noise η	0	0.1	0	0.1	0	0.1	0	0.1
Raw ridge	0.0216	0.0344	0.0750	0.1361	0.0192	0.0681	0.8951	0.8975
Raw RBF	0.0197	0.0323	0.0131	0.0647	0.0129	0.0442	0.8969	0.8994
Boosted trees	0.0201	0.0323	0.0040	0.0586	0.0133	0.0481	0.8975	0.8997
Persistence blend	0.0219	0.0345	0.0750	0.1362	0.0192	0.0684	0.8955	0.8988
Common basis	0.0190	0.0310	0.0251	0.0686	0.0157	0.0460	0.8959	0.8983
KAHM departure (bounded)	0.0192	0.0314	0.0101	0.0622	0.0129	0.0439	0.8972	0.8990
RBF departure (bounded)	0.0189	0.0309	0.0115	0.0634	0.0133	0.0436	0.8968	0.8989
KAHM departure (unbounded)	0.0198	0.0321	0.0108	0.0619	0.0129	0.0440	0.8975	0.8995
RBF departure (unbounded)	0.0190	0.0310	0.0118	0.0627	0.0127	0.0437	0.8972	0.8993
C-selected non-KAHM policy	0.0192	0.0312	0.0040	0.0590	0.0127	0.0434	0.8978	0.8997

The machine-readable files `horizon_metrics.csv` and `condition_horizon_metrics.csv` retain every method, law, noise level and horizon, including raw MSE and the A-only loss scale. They are not pooled with historical losses that use different normalizations. The `replicate_contrasts.csv` file contains all 24 margin-gap values for each primary contrast, allowing the six lower bounds to be recomputed without importing any forecasting model.

B.4 Stability, numerical safeguards and correction activation

The bounded-feature checks cover 192 tasks for each of three bounded procedures. Coefficient and correction checks additionally distinguish the three horizons, giving 576 checks per procedure for each inequality. The independent analyzer checks the recorded fitted-state bound formulas and recomputes actual D corrections as the saved forecast minus the saved raw ridge forecast. It does not claim to reconstruct the fitted coefficient matrices from summary statistics alone. Table 5 reports maxima over the complete roster; maxima in different columns need not come from the same task or horizon.

Table 5: Observed ratios to the deterministic bounded-feature and ridge-readout limits. Feature ratios concern the common block for the common procedure and the respective departure block for the two appended procedures; the readout bound uses the complete concatenated feature radius. A zero correction with a zero bound contributes ratio zero. "Violations" sums failed feature, coefficient and correction checks at the fixed numerical tolerances. These are implementation checks, not confidence intervals for forecast risk.

Bounded procedure	Feature norm / radius	Coefficient norm / bound	Correction / bound	Violations
Common basis	0.99493	0.56103	0.29634	0
KAHM departure (bounded)	0.99943	0.55231	0.20110	0
RBF departure (bounded)	0.99734	0.44963	0.19674	0

The maximum taskwise ratio of KAHM maximum absolute error to ridge maximum absolute error is 1.553, and 0 tasks fail at least one declared stability screen. Local KAHM reconstruction fallback affects 0 A/B/C rows and 0 D rows across all tasks; the corresponding RBF D count is 0. Counts are unions over local models within a task and split, so the same row is not counted again merely because several local maps invoke their guard. The fixed local fallback sets the affected departure to zero. No method fails completely, and no final prediction cell requires recovery. These checks cover the complete task roster.

Table 6: C-selected correction activation. Nonlinear and linear counts have 432 and 144 task/horizon cells, respectively. The four mixing-weight columns combine all 576 cells. A policy choice of raw ridge is counted as zero correction; raw ridge itself has no mixing selection. Nonzero mixing is not an empirical guarantee of D improvement, and no minimum activation frequency was part of the decision.

Procedure	Nonlinear active / 432	Linear active / 144	$\gamma = 0$	$\gamma = 1/4$	$\gamma = 1/2$	$\gamma = 1$
Raw ridge	—	—	—	—	—	—
Raw RBF	427	56	93	52	158	273
Boosted trees	396	48	132	106	175	163
Persistence blend	70	32	474	97	5	0
Common basis	430	56	90	9	66	411
KAHM departure (bounded)	432	54	90	23	79	384
RBF departure (bounded)	431	58	87	14	71	404
KAHM departure (unbounded)	428	51	97	40	107	332
RBF departure (unbounded)	429	61	86	26	75	389
C-selected non-KAHM policy	430	63	83	46	100	347

B.5 Measured costs and reproducibility limits

Each task runs serially in a fresh process with one assigned CPU and one numerical thread. The charged primitive-node costs exclude diagnostic-only rank calculations; full-suite timings, including diagnostics, are retained separately. Node accounting includes model candidates that are actually evaluated, rather than only the winning hyperparameters. Wall and CPU times depend on the recorded machine and software environment. The peak whole-task process RSS is 173.6 MiB; it includes imports, all ten arms, shared diagnostics and evidence arrays, and is not a per-method RAM estimate. Serialized state and owned array buffers omit

Table 7: Whole-procedure choices made by the C-selected non-KAHM policy. Each task contributes one choice across all horizons after each candidate's own horizon-wise C mixing selection. Both KAHM arms are excluded. These are validation choices, not choices made from D errors.

C-selected complete procedure	Nonlinear / 144	Linear / 48
Raw ridge	0	1
Raw RBF	19	6
Boosted trees	51	8
Persistence blend	2	20
Common basis	15	4
RBF departure (bounded)	19	1
RBF departure (unbounded)	38	8

Table 8: Mean charged resources over all 192 equal-sized task suites. Fitting and selection include required A fitting, B selection and C calibration. D prediction covers all 1,664 histories and three horizons. Shared computation nodes are charged once to every procedure requiring them; policy fitting pays the union of its non-KAHM candidates, while its prediction/state follow the selected complete procedure. State denotes serialized selected objects; arrays count their owned NumPy buffers. These are measured implementation costs, not minimum deployment complexity.

Procedure	Fit/select wall (s)	Fit/select CPU (s)	D predict wall (ms)	State (KiB)	Arrays (KiB)
Raw ridge	0.0010	0.0010	0.147	1.10	0.63
Raw RBF	0.0135	0.0135	2.925	65.58	64.28
Boosted trees	0.6452	0.6452	19.046	373.94	335.61
Persistence blend	0.0011	0.0011	0.197	1.15	0.66
Common basis	0.0367	0.0353	5.363	9.00	6.82
KAHM departure (bounded)	0.0833	0.0819	29.245	118.81	114.76
RBF departure (bounded)	0.0658	0.0644	25.758	55.32	51.85
KAHM departure (unbounded)	0.0812	0.0798	26.655	118.81	114.76
RBF departure (unbounded)	0.0642	0.0628	23.810	55.32	51.85
C-selected non-KAHM policy	0.7374	0.7360	15.068	151.36	137.94

transient workspace and do not measure online update latency. The study imposes no application-specific cost ceiling.

The analysis program verifies the prospective source and input checksums, the complete task roster, exact target correspondence, A variance scales, policy predictions, nonfinite/failure accounting and all primary aggregation. The reporting builder separately reconstructs condition and repetition means and all six t lower bounds before producing these displays.

Selected-state inference is compared with the saved forecasts using relative tolerance 10^{-10} and absolute tolerance 10^{-12}. Reproduction at these tolerances requires compatible numerical kernels as well as matching package versions. Processor-dependent vectorized arithmetic can produce differences that exceed the pointwise comparison criterion. The archived forecast arrays define the reported scores. The anonymous supplement reconstructs the losses, repetition aggregates, and six simultaneous confidence bounds directly from those numeric forecasts and the training-target variance scales, without evaluating serialized model states.

Numerical development checks, model replay and independent calculations are consistency checks; they do not constitute a new scientific replication. The protocol was specified locally before evaluation and was informed by the earlier experiments. It was not publicly preregistered. Neither these known-law repetitions nor the algebraic distinctness certificate establishes a natural-application forecasting claim.